Classify Taxonomic Resolution

Description

Classifies taxa in a data frame to a specified taxonomic resolution using a classification table, and aggregates pollen proportions accordingly. Supported resolutions are 'kingdom', 'phylum', 'class', 'order', 'family', 'genus', and 'species'.

Usage

```
classify_taxonomic_resolution(
  data_source,
  data_classification_table,
  vec_taxonomic_resolution,
  verbose = TRUE
)
```

Arguments

`data_source`	A data frame containing taxon data with columns including 'taxon', 'dataset_name', 'age', and 'value'.
`data_classification_table`	A data frame mapping 'sel_name' to taxonomic levels. Must contain at least one rank column at or below 'vec_taxonomic_resolution' (e.g. 'family', 'genus', 'species').
`vec_taxonomic_resolution`	A character string specifying the finest taxonomic level to use. Must be one of "kingdom'‘, ''phylum'‘, ''class'‘, ''order", "family'‘, ''genus'‘, or ''species". Taxa will be classified at this rank if possible, or at the coarsest available rank below it if not (fallback behaviour).
`verbose`	Logical. If 'TRUE' (default), classification warnings and fallback messages are printed.

Details

Performs a left join to map taxa to all available rank columns up to and including 'vec_taxonomic_resolution'. The finest non-NA rank is then selected via 'dplyr::coalesce()' applied from finest to coarsest. This means a taxon known only to family when genus is requested will be assigned to its family name rather than dropped.

Taxa with no valid classification at any available rank are removed with a 'cli::cli_warn()' warning. Taxa that fall back to a coarser rank are reported via 'cli::cli_inform()'. Ranks finer than 'vec_taxonomic_resolution' (e.g. species when genus is requested) are never used, even when present in the classification table. The NA-drop step prevents a column literally named NA appearing in the community matrix produced by downstream 'pivot_wider()' calls.

Value

A data frame with taxa classified to the finest available rank at or below 'vec_taxonomic_resolution' and pollen proportions aggregated accordingly. The output preserves all dataset_name and age combinations for true negatives.

See Also

[filter_community_to_plantae()], [filter_community_by_minimum_proportion()]